

Project Overview: Student Data Analysis & Automation Pipeline

Team 4 Analysis

Strategic Insight, Data-Driven Execution

Executive Summary

Objective: Automate the processing of the "RIT Opportunity Wise" student dataset to extract actionable insights.

Scope: Analyzed 8,558 unique student entries across multiple engagement categories.

Methodology:

- **Pipeline:** Developed a two-step Python workflow utilizing `pandas` for statistical analysis and `matplotlib/seaborn` for high-fidelity visualization.
- **Outcomes:** Automated generation of a comprehensive text report and a suite of high-resolution charts.

Key Data Insights

- **Dominant Engagement:** Internships are the primary driver of student participation, accounting for 63.34% of the dataset.
- **Geographic Concentration:** Activity is heavily concentrated in the United States (46.46%) and India (33.14%).
- **Demographics:** Participant pool is 58.64% Male and 41.15% Female.
- **Conversion Pipeline:**
 - **Team Allocated:** 38.28% (Active cohort).
 - **Rejected:** 41.70% (Indicates high competitive barrier).
 - **Dropped Out:** 7.21% (Attrition metric to monitor).

Automation Technical Architecture :-

Reporting Engine (Step 1)

- Uses `pandas` for automated data validation and file checking.
- Generates `summary_report.txt` with formatted percentage distributions and popular program rankings.

Visualization Pipeline (Step 2)

- Uses `seaborn` and `matplotlib` for standardized visual storytelling.
- **Output:** Three high-density (300 DPI) images:

- `opportunity_categories.png`
- `top_countries.png`
- `status_distribution.png`

Conclusion & Next Steps :-

- **Reproducibility:** The Python scripts ensure consistent reporting regardless of dataset growth.
- **Operational Efficiency:** The automated pipeline replaces manual data tabulation, reducing potential for human error.
- **Future Focus:** Monitor the "Dropped Out" segment (7.21%) to improve retention and analyze the "Rejected" cohort to optimize future admission criteria.

STEP 1:- PRINT OUT TEXT-TABLES AND PERCANTAGE NEEDED FOR THE REPORT.

Python Script :

```
import pandas as pd
import os

# Try to find the file
filename = 'RIT+Opportunity+Wise+Data+--+Sheet1.csv'

if not os.path.exists(filename):
    print(f"ERROR: Cannot find the file '{filename}'!")
    print("Please make sure you save this python file in the EXACT same folder where your data file is.")
else:
    df = pd.read_csv(filename)

# Open a clean text file to write the results into
with open("summary_report.txt", "w") as f:
    f.write("=== TASK 4: SUMMARY STATISTICS REPORT ===\n")
    f.write(f"Total Rows Analyzed: {len(df)}\n\n")

    f.write("=== 1. OPPORTUNITY CATEGORY BREAKDOWN ===\n")
    cat_counts = df['Opportunity Category'].value_counts()
    cat_percentages = df['Opportunity Category'].value_counts(normalize=True) * 100
    for cat in cat_counts.index:
        f.write(f"{cat}: {cat_counts[cat]} students ({cat_percentages[cat]:.2f}%)\n")

    f.write("\n=== 2. TOP 5 MOST POPULAR PROGRAMS ===\n")
    top_programs = df['Opportunity Name'].value_counts().head(5)
```

```

for prog, count in top_programs.items():
    f.write(f"- {prog}: {count} students\n")

f.write("\n=== 3. GEOGRAPHIC DISTRIBUTION (TOP 5 COUNTRIES) ===\n")
country_counts = df['Country'].value_counts().head(5)
country_percentages = df['Country'].value_counts(normalize=True).head(5) * 100
for country in country_counts.index:
    f.write(f"{country}: {country_counts[country]} students\n")
    f.write(f"({country_percentages[country]:.2f}%)")

print("SUCCESS! A clean report has been created.")
print("Look in your folder for a new file called: 'summary_report.txt'")

```

STEP 2:- CHARTS GENERATION AND EMBEDDATION IN THE REPORT.

Python Script :

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
df = pd.read_csv('RIT+Opportunity+Wise+Data+-+Sheet1.csv')

# Set a clean, modern aesthetic style for the charts
sns.set_theme(style="whitegrid")

print("Generating charts... Please wait...")

# CHART 1: Opportunity Categories (Horizontal Bar Chart)
plt.figure(figsize=(9, 4.5))
cat_data = df['Opportunity Category'].value_counts()
sns.barplot(x=cat_data.values, y=cat_data.index, palette="Blues_r")
plt.title("Student Registrations by Opportunity Category", fontsize=14, fontweight='bold')
plt.xlabel("Number of Registrations", fontsize=11)
plt.ylabel("Category", fontsize=11)
plt.tight_layout()
plt.savefig("opportunity_categories.png", dpi=300) # Saves image to your folder
plt.close()
print("- Saved opportunity_categories.png")

```

```
# CHART 2: Top 5 Countries
plt.figure(figsize=(9, 4.5))
country_data = df['Country'].value_counts().head(5)
sns.barplot(x=country_data.values, y=country_data.index, palette="Purples_r")
plt.title("Top 5 Countries by Student Participation", fontsize=14, fontweight='bold')
plt.xlabel("Number of Sign-ups", fontsize=11)
plt.ylabel("Country", fontsize=11)
plt.tight_layout()
plt.savefig("top_countries.png", dpi=300)
plt.close()
print("- Saved top_countries.png")

# CHART 3: Engagement Funnel Statuses
plt.figure(figsize=(9, 5))
status_data = df['Status Description'].value_counts()
sns.barplot(x=status_data.values, y=status_data.index, palette="Reds_r")
plt.title("Distribution of Student Application Statuses", fontsize=14, fontweight='bold')
plt.xlabel("Number of Students", fontsize=11)
plt.ylabel("Status", fontsize=11)
plt.tight_layout()
plt.savefig("status_distribution.png", dpi=300)
plt.close()
print("- Saved status_distribution.png")

print("\nAll charts successfully created and saved to your project folder!")
```
